



MOHYEDDINE TALEB

PhD in Biology, Chemistry, and Health

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Profil

PhD in Biology, Chemistry, and Health with strong expertise in processing complex data, computational modeling, and bioinformatics. Passionate about Computational Biology, Data Science, and Artificial Intelligence, I possess advanced skills in Python programming, machine learning, and statistical analysis. Accustomed to managing massive databases via high-performance computing (HPC), I apply my rigor and adaptability to solve complex technical and biological challenges.

Technical Skills

- **Data Science & AI:** Python (Pandas, NumPy, Scikit-Learn, MDAnalysis), R, PyTorch, Machine Learning, Predictive Modeling, Data Visualization, Statistical Analysis, Big Data processing.
- **Tools & Software Engineering:** Git, Databases (SQL), High-Performance Computing (HPC: ROMEO, CRIANN, SLURM, Bash), Data pipeline automation, Algorithms, Agile methods.
- **Bioinformatics & Modeling:** Molecular Dynamics and Docking (GROMACS, AutoDock, Vina), 3D Modeling (ChimeraX, PyMOL), Quantum Chemistry (ORCA, Gaussian). NGS and Sequence Analysis (BLAST, Clustal Omega, DESeq2, edgeR, Biostrings).
- **Experimental Sciences:** Biochemistry, protein production and purification (*E. coli*), Spectroscopy (NMR, UV-Vis, MS), Chromatography (HPLC).
- **Languages:** English (Fluent/Professional), French (Fluent), Arabic (Native).

Professional Experiences

Doctoral Research – Biochemistry & Bioinformatics

October 2022 – December 2025 | ICMR – CNRS 7312, University of Reims, France

- Developed and optimized Python scripts for the extraction, processing, and statistical analysis of massive data from molecular simulations.
- Analyzed biological sequences (alignment, polymorphism identification) prior to structural modeling.
- Designed and automated data pipelines (workflows) for molecular docking execution and density analysis using IGMPlot.
- Executed complex calculations and managed databases on supercomputers (HPC - Linux/SLURM environments).
- Conducted predictive modeling and molecular geometry optimization.
- Experimental phase: Optimized protocols for the production and purification of the TSPO transmembrane protein; conducted quantitative and qualitative analyses.

Research Engineer (Master 2 Internship)

January 2022 – June 2022 | LRB-Inserm 1039 Laboratory, Grenoble, France

- Synthesized and evaluated the biological activity of flavone analogs.
- Analyzed spectral (NMR, mass spectrometry) and chromatographic data.

Environmental Control Engineer

January 2021 – June 2021 | HUPP Company, Constantine, Algeria

- Collected and analyzed data for environmental impact assessments.
- Implemented quality controls and drafted compliance reports.

Intern Engineer (Master 2 Thesis)

January 2020 – June 2020 | Higher National School of Biotechnology, Constantine, Algeria

- Extracted and purified essential oils (*Physalis L.*) and analyzed their biological activity.

Education and Diplomas

PhD in Biology, Chemistry, and Health (Modeling & Bioinformatics)

2022 – 2025 | University of Reims Champagne-Ardenne (URCA), France (Thesis defended December 15, 2025)

- Project: INTERDY - Interaction and molecular dynamics studies (Dir. Dr. Luminita Duma).

International Master (MSc) in Bio-Health Engineering

2021 – 2022 | Université Grenoble Alpes (UGA), France

Master in Pharmaceutical Biotechnology & Engineering in Biotechnology

2015 – 2020 | École nationale de Biotechnologie (ENSB), Constantine, Algérie

Communications and Interests

- **Oral Communications:** PhDay (English, 2024, 2025); ROMEO - HPC Days (French, 2024, 2025); 16th Meetings of the Grand Est Theoretical Chemistry Network (English, 2025); Appicom 4th plenary meeting (English, 2023).
- **Certifications & Training:** Design Thinking (Business plans), Teamwork and Leadership.
- **Design & UX:** UX Consultant (Freelance), User Research and Testing, Proficiency in Photoshop / Illustrator.
- **Hobbies:** Gym, Football, Hiking.

Publications

- *Isolation, in vitro acetylcholinesterase inhibitory activity, molecular docking, and in silico ADME analysis of compounds from Cadaba farinosa roots – (Published, DOI: 10.1016/j.phytol.2026.104254).*
- A primary research article detailing the findings of my PhD thesis (submitted).
- **Ongoing Research:** Currently collaborating with researcher Tobias Koch (Germany) and CIRAD lab in Montpellier on a theoretical chemistry project investigating the molecular diffusion of aroma molecules in lipid membranes using umbrella sampling and quantum chemistry optimization (ORCA).